

Python, NumPy & AI Interview Questions with Answers

1. What is Python?

Python is a high-level, interpreted programming language used in AI, web development, automation, and data science.

2. What is NumPy?

NumPy is a Python library used for numerical computing and array operations.

3. What is an API?

API stands for Application Programming Interface. It allows communication between applications.

4. Why is NumPy faster than Python lists?

NumPy uses optimized C-based operations and less memory.

5. Difference between List and Tuple?

List is mutable while Tuple is immutable.

6. What is Flask?

Flask is a lightweight Python web framework used to build APIs and web applications.

7. What is Pinecone?

Pinecone is a vector database used for similarity search in AI applications.

8. What is Broadcasting in NumPy?

Broadcasting allows operations on arrays of different shapes.

9. What is OOP?

OOP stands for Object-Oriented Programming. It includes class, object, inheritance, and polymorphism.

10. What is Exception Handling?

Exception handling manages runtime errors using try and except blocks.

11. What is REST API?

REST API is used for communication between frontend and backend using HTTP methods.

12. What is Vectorization?

Vectorization performs operations on arrays without loops for better performance.

13. What is OAuth 2.0?

OAuth 2.0 is a secure authentication protocol used for API authorization.

14. What is Reshape in NumPy?

`reshape()` changes dimensions of arrays without changing data.

15. What is a Dictionary?

Dictionary stores data in key-value pairs.

16. What is Docker?

Docker is a containerization platform used to deploy applications consistently.

17. What is Lambda Function?

Lambda is an anonymous one-line function.

18. What is AI Resume Matching?

It compares resumes and job descriptions using embeddings and vector similarity search.

19. What is a Generator in Python?

Generators use yield keyword to return values one by one.

20. Why is Python popular in AI?

Because of easy syntax, powerful libraries, and strong community support.

21. What is Ngrok?

Ngrok creates secure public URLs for local servers.

22. Difference between append() and extend()?

append() adds one item while extend() adds multiple items.

23. What is File Handling?

File handling is used to read and write files in Python.

24. What are Webhooks?

Webhooks automatically send data between applications when events occur.

25. What is Multithreading?

Multithreading allows multiple threads to execute simultaneously.

26. What is JSON?

JSON is a lightweight format for data exchange between applications.

27. What is a Virtual Environment?

It isolates project dependencies separately.

28. What is PEP 8?

PEP 8 is the coding style guide for Python.

29. What is an Iterator?

Iterator accesses elements one at a time.

30. What is Broadcasting in NumPy?

It allows NumPy arrays of different sizes to work together.